Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

SDC HACKATHON 2026

Phase 1: Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

1. Team & Project Overview

Team Name	16.0
Selected Domain	Aiversecity - Future-Ready Workforce
Selected SDG Track (1–17)	SDG 4 - Quality Education (related: SDG 8 - Decent Work and Economic Growth)
Project / Solution Name	ShiftHappens
Name (Leader)	**WONG PYNVEN**
Email & Phone (Leader)	**pynven323@gmail.com & 016-3266323**
University Affiliation(s)	**University of Malaya**
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

AI is changing office work faster than workers can adjust. TalentCorp's 2024 impact study estimates that about 620,000 jobs - 18% of 3.5 million employees in ten key Malaysian sectors - are highly impacted by AI, digitalisation and the green economy within three to five years [1].

The people most exposed are mid-career white-collar workers in routine-heavy jobs: administrative and clerical staff, data entry clerks, customer service agents, accounting and payroll support, and junior content and marketing staff. Their moment of need comes when AI tools arrive at work, or when the news says their job is "at risk".

At that moment they face four problems. First, risk is described by job title, so they cannot tell which of their daily tasks will change. Second, they get generic course lists instead of a clear next skill and role. Third, they have little time and money because they already work full time. Fourth, they distrust black-box AI advice and worry about uploading a CV full of personal data.

This matters socially and economically. Workers who cannot see the change coming reskill too late, which raises the risk of job loss and income insecurity. Helping these workers act early supports SDG 4.4 (relevant skills for decent jobs) and SDG 8.5 (productive employment for all).

[1] TalentCorp impact study on AI, digital and green economy, Nov 2024 (reported by The Edge Malaysia, Malay Mail).

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

ShiftHappens is a deployed web app that takes a worker from "will AI take my job?" to "which role should I aim for, and what do I learn next?". Its core idea: AI changes tasks, not whole jobs.

A five-step journey

1. Your background - upload or paste a CV, pick one of five common office roles, or re-upload a saved profile.

2. Review - the user confirms tasks, skills and levels; the AI's suggestions are shown read-only with evidence and confidence.

3. Your day - each daily task is sorted into "AI may take over", "AI will assist you" or "stays mostly human", using a dated, updatable trend-signal layer under two scenarios (steady vs rapid AI progress).

4. Your roles - ten ESCO future roles ranked by an explainable future-fit score (skill readiness plus market outlook, weighted by the user), with essential-skill gaps, SDG tags and a 14-case fairness check.

5. Your plan - an ordered plan of free or low-cost, hand-verified courses with weeks and cost, fitted to the user's hours, budget and timeline.

How AI is applied

Google Gemini reads the CV - after emails, phone numbers and IC numbers are removed - and returns structured JSON (skills with level, evidence and confidence, plus daily tasks), which is mapped to the ESCO skills taxonomy. Trend signals then drive task baskets, skill outlooks and role ranking through fixed formulas, so every score can be explained and the system cannot invent jobs or courses.

Workshop skills we use

GenAI APIs: Gemini with structured JSON output and keys kept in secrets. Data architecture: versioned, schema-validated JSON datasets (trend batches, ESCO skills and roles, course catalogue) with no personal data stored. Cloud hosting: a live Replit deployment.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Technologies

- Language and app framework: Python; Streamlit 1.66.1 (UI and server); custom CSS and JavaScript for the intro animation, step dock and command palette.

- GenAI API: Google Gemini API via the google-generativeai SDK; model set by LLM_MODEL (currently gemini-3.8-flash).

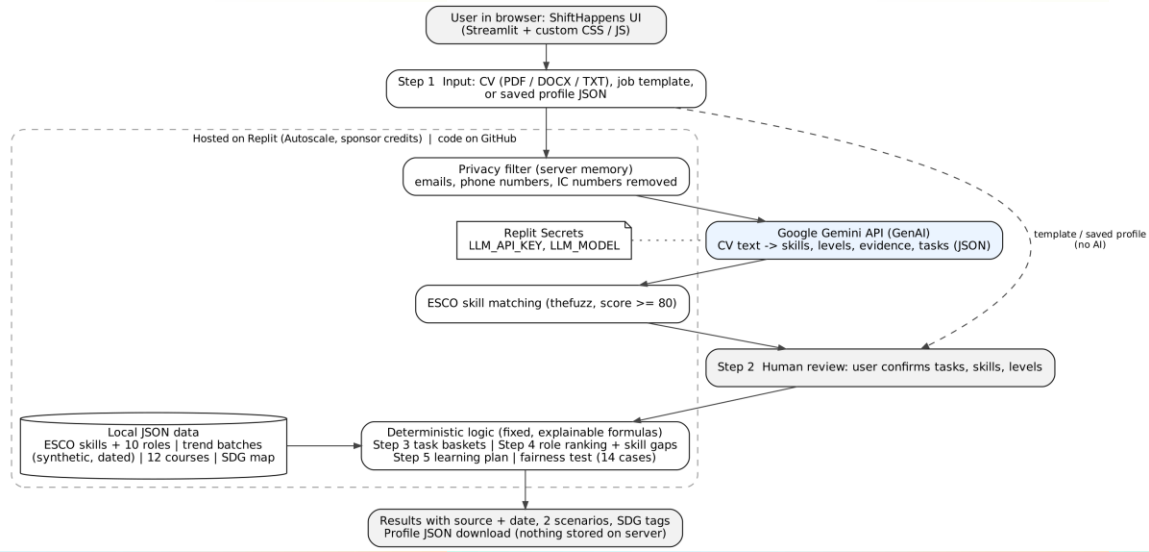
- Document parsing and matching: PyPDF2, python-docx, thefuzz (fuzzy matching to ESCO), pandas.

- Data: JSON files - ESCO v1.2.1 skills and 10 future roles, 5 job templates, synthetic trend batches (2026-08 to 2026-10), 12 verified courses, SDG mapping, skill-to-task map. A built-in validator checks every trend batch.

- Database: none in Phase 1, by design - no personal data is stored; users keep their own profile file.

- Hosting and tools: Replit Autoscale deployment with Replit Secrets for the API key; Git and GitHub.

System block diagram



Sponsor-provided GenAI API or cloud platform

Yes. Replit (cloud platform): we used USD 25 of sponsor-provided Replit credits to deploy the live prototype at https://sdc-hackathon-2026-team-160.replit.app (Autoscale deployment, API key in Replit Secrets). The Gemini API is not sponsor-provided; we use our own free-tier key. RunPod was not used.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Empathise and define: from the domain brief and the TalentCorp study, we narrowed the users to one group - mid-career office workers in routine-heavy roles - and one moment of need: hearing that AI may replace their job.

Ideate: we reframed the question from "is my job at risk?" to "which of my tasks change, and what next?", which led to task baskets, two scenarios and a user-controlled trend weight.

Prototype and test: we built in small, working increments and tested every change end to end with fictional CVs and job templates. Testing changed the design: a results page became a five-step guided journey, a human review step was added before any matching, and we added source and date labels, a fairness check and a saved-profile option.

Deployment: the app runs live on Replit from the GitHub repository, with secrets outside the code and no database to manage. For Phase 2, real job-market data can replace the synthetic batches using the same validated file format.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Primary - Target 4.4: "By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship." Secondary - Target 8.5: "By 2030, achieve full and productive employment and decent work for all women and men..." Source: UN SDGs, https://sdgs.un.org/goals
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Problem: about 620,000 jobs (18% of 3.5 million employees in 10 key Malaysian sectors) are highly impacted by AI, digitalisation and the green economy within 3 to 5 years (TalentCorp, Nov 2024). Product starting point: today only 11.5% of the essential skills of our 10 future roles (34 of 296) are covered by a verified free course, so most users get only a partial learning plan (team calculation from the app's data files, 25 Oct 2026).
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Primary KPI: % of essential skills across our 10 future roles that have at least one verified free or low-cost course (this decides how much of each user's skill gap the plan can actually close).

Metric Category	Your Response
	Supporting KPI: minutes for a new user to go from profile to a ranked role list and a learning plan.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Raise verified course coverage of essential skills from 11.5% to at least 40% by Phase 2, and keep the profile-to-plan journey under 5 minutes for a new user.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [x] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- [x] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- [x] The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- [x] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☒ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

Not applicable.

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☒ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Streamlit 1.66.1	Code (framework)	Apache License 2.0
pandas	Code (library)	BSD 3-Clause
google-generativeai (Gemini SDK)	Code (library)	Apache License 2.0
thefuzz	Code (library)	MIT
python-docx	Code (library)	MIT
PyPDF2	Code (library)	BSD 3-Clause
python-dotenv	Code (library)	BSD 3-Clause
Geist font (Vercel)	Code (font)	SIL Open Font License 1.1
Material Symbols (bundled with Streamlit)	Code (icons)	Apache License 2.0

Google Gemini (gemini-3.8-flash) via Gemini API	Model (pre-trained, used through API)	Gemini API Additional Terms of Service
ESCO v1.2.1 skills, occupations and relations (European Commission)	Dataset (public)	Free reuse with attribution, European Commission reuse policy
Course titles and links (Coursera, Khan Academy, HubSpot Academy, Kaggle Learn, edX)	Dataset (public course information, links only)	Content owned by each provider; only links and basic facts stored
UN SDG goals and targets text	Dataset (public)	United Nations public information

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☒ No

If yes, please describe the nature of the agreement and which member it concerns:

Not applicable.

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



Date: 25 October 2026